

QUESTION: 20 GROUP: Aptitude SECTION: Problem Solving
Mark(s): 1

Out of $2n + 1$ students, n students have to be given the scholarships. The number of ways in which at least one student can be given the scholarship is 63. What is the number of students receiving the scholarship?

☐ 7

☐ 9

☒ 3

☐ 5

I worked my way up from the options... Check for the third option which will either give the answer or will eliminate three options for you.

C option- if n is 4

if n is 4
Then $2n+1$ is 9

And all selecting atleast one candidate comes out to be $(9C1 + 9C2 + 9C3 + 9C4)$ which we can see will go way beyond 63.

So this options eliminate any value of n equal to or greater than 4. So option C/D/E are out.

Option B - If n is 3

If $n=3$
 $2n+1=7$

And all selecting atleast one candidate comes out to be $(7C1 + 7C2 + 7C3)$ which is 63.

So CORRECT answer is B

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Directions: The question is followed by two statements (I) and (II). Answer the question using the following instructions.

Mark (N) if the question can be answered by one of the statements alone but not by the other.

Mark (H) if the question can be answered by using either statement alone.

Mark (I) if the question can be answered using both statements together, but can not be answered using either statement alone.

Mark (U) if the question cannot be answered.

What is the height of 'R'?

- I: The average height of 'P' and 'S' is equal to average height of 'S' and 'R'.
- II: P's height is five feet.

☒ (D)

☐ (B)

☐ (A)

☐ (C)

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QUESTION 12 GROUP: Aptitude SECTION: Puzzles
Mark(s): 1

If "AQUARIUMS" is coded as "CCKOSTUWW", then what is the code for "BARBIEQUE"?

☐ DCGKSTWDG

☐ DCTDKGSWG

☐ CDDGCKSTW

☐ CDGKSTWDG

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QUESTION: 10 GROUP: Aptitude SECTION: Problem Solving

Mark(s): 1

TIME LEFT: 00:28:58

Mr. S purchased a number of articles, all at a discount of $100/3\%$ of the list price. However he is required to pay a duty of 20% on his cost of goods. If he realizes a profit of 25% on his outlay, what must he charge for an article priced Rs. 228 as the list price?

☐ Rs. 348

☐ Rs. 332

☒ Rs. 228

☐ Rs. 240

Answer:

228

Step-by-step explanation:

Price of article after discount = $228 - 228 \times (1/3) = 152$

Cost including duty of 20% = $152 + (152 \times 20/100) = 912/5$

25% profit:

$(912/5) + (25/100) \times (912/5) = 912/4 = 228$

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QUESTION 7 GROUP: Aptitude SECTION: Puzzles

Marked: 1

00:29:19

How many pairs of letters are there in the word 'ADDICABLE' which has as many letters between them (from both the sides) as in the alphabets from A to Z?

☐ More than 3

☐ 0

☒ 2

☐ 3

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QUESTION: 9 GROUP: Aptitude SECTION: Problem Solving

Mark(s): 1

Ajit and Vikas started a joint venture. If Ajit invested Rs. 40000 in the beginning and at the end of the 2 months he added another Rs. 20000. Vikas withdraw an amount of Rs. 15000 after 4 months. If both Ajit and Vikas are sharing the profits equally at the end of the year then find the approximate amount invested by Vikas initially.

☐ Rs. 46666

☐ Rs. 57666

☒ Rs. 66666

☐ Rs. 53333

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QUESTION 6 GROUP: Aptitude SECTION: Problem Solving

Mark(s) 1

A cricket batsman scored 200 runs in which there were 4 sixes and 5 fours while the remaining score was by running between the wickets. What percentage of his total score did he make by running between the wickets?

☐ 43%☐ 72%☐ 56%☒ 78%[Previous Question](#)[Next Question](#)

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QUESTION: 5 GROUP: Aptitude SECTION: Problem Solving

Mark(s): 1

Kapil scored an average of 60 points in five rounds of a game. In sixth round he scored only 30 points. What is the average score of Kapil after sixth round?

☒ 58☐ 55☐ 54☐ 57[Previous Question](#)[Next Question](#)

DELL

Mark(s) 1

Direction: The question is followed by two statements (I) and (II). Answer the question using the following instructions:

Mark (A) if the question can be answered by one of the statements alone but not by the other

Mark (B) if the question can be answered by using either statement alone

Mark (C) if the question can be answered using both statements together, but cannot be answered using either statements alone

Mark (D) if the question cannot be answered even by using both the statements together

Is y divisible by 26?

(I) When y is divided by 13, it leaves 6 as the remainder.

(II) When y is divided by 39, it leaves 20 as the remainder.

☐ (D)

☐ (A)

☐ (C)

☐ (B)

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DELL

A goods train traveling from station A to B meets with an accident one hour after starting. After a stoppage of 30 minutes there, due to damage it now proceeds at 80% of normal speed and arrives 2 hours late. Had the accident happened 80 km further on, it would have reached one - and - a - half hours earlier without any stoppage. What is the original speed of the train?

☐ 25 km/hr

☐ 22.5 km/hr

☒ 20 km/hr

☐ 30 km/hr

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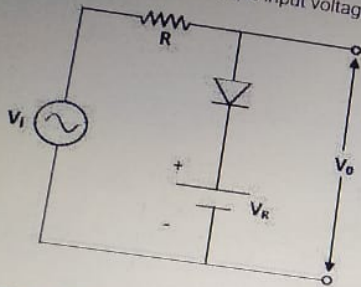
DELL

QUESTION: 48 GROUP: Hardware SECTION: Analog Circuits

Mark(s): 1

TIME LEFT 00:40:45

In the following circuit, V_I is input voltage, V_O is output voltage and V_R is a reference voltage. If V_I is greater than V_R , then _____



☐ $V_O = V_I - V_R$

☐ $V_O = V_I + V_R$

☐ $V_O = V_I$

☒ $V_O = V_R$

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DELL

QUESTION: 50

GROUP: Hardware

SECTION: Digital Electronics

Mark(s): 1

The excitation table for the flip-flop is given below. Mention the name of the flip-flop.

Q	Q+1	A	B
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

☐ J-K Master slave☐ D☐ SR☐ J-K

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Instructions

QUESTION: 49

GROUP: Hardware

SECTION: Semiconductor Physics

TIME LEFT 00:40:39

Mark(s): 1

At what temperature, a germanium semiconductor having energy gap of 0.72 eV will become conductive if electron-hole-pair generation takes place only when the operating temperature is equal or greater than the temperature corresponding to energy gap.

☐ 53000 °C☐ 53 °C☐ 1300 °C☐ 5300 °C

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DELL

QUESTION 46 GROUP: Hardware SECTION: RTL

Mark(s): 1

A digital RTL system working in synchronously operating mode means?

- ☐ All units (Processing unit (PU) and control unit (CU)) work at a same high level (Level 1)
- ☐ All units (Processing unit (PU) and control unit (CU)) work differently at a same clock pulse.
- ☐ All units (Processing unit (PU) and control unit (CU)) work simultaneously at a same clock pulse.
- ☐ All units (Processing unit (PU) and control unit (CU)) work at a same Low level (Level 0)

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QUESTION: 47 GROUP: Hardware SECTION: Digital Electronics

Mark(s): 1

A mod 50 counter can be implemented using:

- ☐ 10 mod-5 counters
- ☐ 5 mod-10 counters
- ☐ Mod-10 counter followed by mod-5 counter
- ☐ 10 mod-10 counters

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QUESTION: 45 GROUP: Hardware SECTION: Analog Circuits

Mark(s): 1

For an depletion MOSFET the pinch voltage is found to be -6V . If the current I_D and I_{DSS} are 9mA and 16mA respectively. What is the gate source voltage?

☐ -1.5V

☐ 1.5V

☐ -6V

☐ 6V

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QUESTION 46 GROUP: Hardware SECTION: RTL

Mark(s): 1

A digital RTL system working in synchronously operating mode means?

- ☐ All units (Processing unit (PU) and control unit (CU)) work at a same high level (Level 1).
- ☐ All units (Processing unit (PU) and control unit (CU)) work differently at a same clock pulse.
- ☐ All units (Processing unit (PU) and control unit (CU)) work simultaneously at a same clock pulse.
- ☐ All units (Processing unit (PU) and control unit (CU)) work at a same Low level (Level 0).

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QUESTION: 42 GROUP: Hardware SECTION: VLSI

Mark(s): 1

For NAND gate, Determine the value of propagation delay for high to low (t_{phl}), if the value of propagation delay for low to high (t_{plh}) is 12ns and total propagation delay is 10ns-

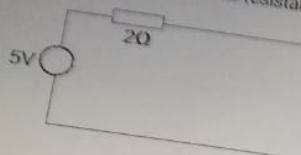
☐ 10ns☐ 6ns☒ 8ns☐ 12ns[Previous Question](#)[Next Question](#)

DELL

QUESTION: 40 GROUP: Hardware SECTION: Basic Electronics

Mark(s): 1

The following voltage source is connected to a 2Ω resistance in series. If this source is converted to an equivalent current source, what will be the value of the current source and its resistance?



- ☐ 10A with a series resistance of 2Ω
- ☐ 10A with a parallel resistance of 5Ω
- ☐ 5A with a series resistance of 2Ω
- ☐ 10A with a parallel resistance of 2Ω

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QUESTION: 43 GROUP: Hardware SECTION: VLSI
Mark(s): 1

09:41:24

Which of the following process technology is not used in FPGA?

☐ EEPROM

☐ EPROM

☐ Anti Fuse

☐ SRAM

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Instructions

QUESTION: 39

GROUP: Hardware

SECTION: Microprocessors and Microcontrollers

Mark(s): 1

TIME LEFT: 00:42:11

in terms of structure, differentiate the microprocessor with respect to microcontroller.

- ☐ Microprocessor is a part or unit of Microcontroller.
- ☐ Function of Microcontroller & microprocessor is same for a same device
- ☐ Microcontroller is a part of microprocessor
- ☐ Function of Microcontroller & microprocessor is totally different

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Mark(s): 1

What is the result of the given program?

```

program messagecheck();
    vmm_log log = new("test_log", "log");
    int error_cnt;

    initial begin
        `vmm_note(log, $psprintf("Error Message at time %d", $time));
        `vmm_error(log, " ERROR Message A ");
        `vmm_error(log, " ERROR Message B ");

        error_cnt = log.get_message_count(vmm_log::ERROR_SEV, "/", "/", 1);
        $display(" Total Number of Error messages : %0d ", error_cnt);
    end
endprogram
    
```

☐ Total Number of Error messages: 1

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```
error_cnt = log.get_message_count(vmm_log::ERROR_SEV, "/./", "/./", 1);  
$display(" Total Number of Error messages : %0d ",error_cnt);  
end  
endprogram
```

TIME LEFT 00:4

☐ Total Number of Error messages: 1

☐ Total Number of Error messages: 2

☐ ERROR Message A
ERROR Message B
Total Number of Error messages: 1

☐ ERROR Message A
ERROR Message B
Total Number of Error messages: 2

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QUESTION: 33 GROUP: Hardware SECTION: Digital Electronics

Mark(s): 1

If an analog input signal voltage ranges from 0 to 5 V, then the 8-bit digital encoding for 3.5 V using successive approximation method is

☐ 10100001

☐ 10011100

☐ 10100011

☒ 10110011

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Instructions

QUESTION: 31 GROUP: Hardware SECTION: Digital Electronics

Mark(s): 1

TIME LEFT 00:43:07

Simplify the Boolean expression:

$$Z = A'BCD + AB'C'D' + AB'C'D + AB'CD' + AB'CD + ABC'D' + ABC'D + ABCD' + ABCD$$

☐ A+BCD

☐ ABC+A'C'D

☐ A'B+C'D

☐ A'B'C+BC'D

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QUESTION 32

GROUP: Hardware

SECTION: Digital Electronics

Mark(s): 1

00:42:58

A 4-bit left shift register shifts by 1 bit for every clock pulse. The higher two bits are ORed and the output is fed to a T FF and the output of T FF is fed to the LSB of the initial contents of the register are 1011, what are the contents of the register after 4 clock pulses? Initially the FF is reset.

☐ 1 1 1 1☐ 0 0 0 1☒ 1 0 1 0☐ 1 0 1 1[Previous Question](#)[Next Question](#)

DELL

QUESTION: 30 GROUP: Hardware SECTION: Basic Electronics

Mark(s): 1

Time: 00:43:14

Thevenin's theorem cannot be applied to which of the following circuit?

☐ Circuit with only capacitor

☐ Non linear

☐ Circuit with only inductor

☐ Linear

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QUESTION 29

GROUP: Hardware

SECTION

Microprocessors and Microcontrollers

Mark(s) 1

What is the content of register B after the following set of instructions?

MVI A, 12H

RLC

RLC

RLC

MOV B, A

(Initially carry is clear)

☐ 96 H☐ 36 H☐ 144 H☒ 90 H

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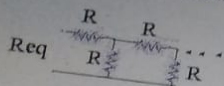
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Instructions

QUESTION: 28 GROUP: Hardware SECTION: Basic Electronics

Mark(s): 1

Determine the equivalent resistance of infinite network shown.



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☐ $2.81R \Omega$

☐ $1.09R \Omega$

☐ $3.16R \Omega$

☒ $1.62R \Omega$

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QUESTION: 27

Mark(s): 1

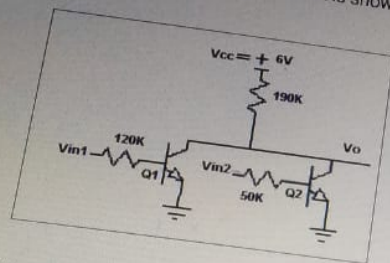
GROUP: Hardware

SECTION: RTL

Instructions

Time 00:43:47

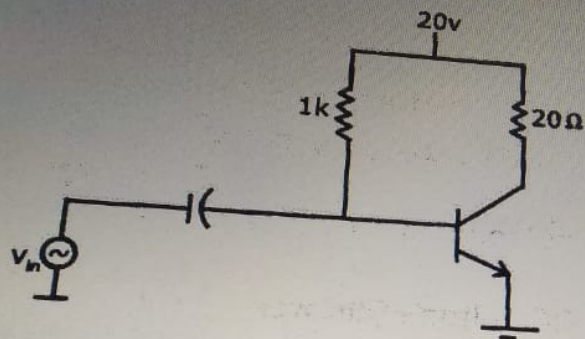
A working diagram of RTL circuit is shown in figure,



Find the level of out put when both inputs are at high level?

- ☐ $V_o = 6V$
 - ☐ $V_o = -6V$
 - ☐ $V_o = 1V$
 - ☒ $V_o = 0V$
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- ☐ Normal class AB amplifier
- ☐ Normal class B amplifier
- ☐ Tuned class A amplifier
- ☐ Normal class A amplifier

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QUESTION: 25 GROUP: Hardware SECTION: RTL
Mark(s): 1

The geometric layout of RTL design is a result of which process?

☐ Synthesis and optimization

☐ Verification

☐ Simulation

☒ Placing & Routing

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QUESTION: 23 GROUP: Hardware SECTION: Semiconductor Physics

Mark(s): 1

TIME LEFT: 00:44:36

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For an extrinsic semiconductor, find the concentration of holes and electrons in a p-type Ge at 300 K if the conductivity is 200 ohm-cm^{-1} . Consider $n_i = 3.5 \times 10^{13} / \text{cm}^3$, and mobility of hole is $1500 \text{ cm}^2/\text{V-sec}$.

☐ $8.33 \times 10^{17} / \text{cm}^3$ & $1.47 \times 10^5 / \text{cm}^3$

☐ $4.10 \times 10^{14} / \text{cm}^3$ & $3.40 \times 10^5 / \text{cm}^3$

☐ $7.36 \times 10^{18} / \text{cm}^3$ & $5.97 \times 10^5 / \text{cm}^3$

☐ $2.55 \times 10^{10} / \text{cm}^3$ & $2.27 \times 10^5 / \text{cm}^3$

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DELL

```
x = (y = (z = 10));  
$display(" x is %d y is %d z is %d ", x, y, z);  
endif  
end  
endmodule
```

☐ value of x is x
x is x y is y z is z

☐ value of x is 2
x is y y is z z is 10

☐ value of x is 4
x is 10 y is 10 z is 10

☐ value of x is 2
x is 2 y is 4 z is 6

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DELL

What is the result of given program?

00:44:21

```
module code {  
  int x,y,z;  
  initial begin  
    x = 2; y = 4; z = 6;  
    if (x > y) {  
      $display("value of x is %d", x);  
      x = (x + (z - 1)) % 10;  
      $display("x is %d y is %d z is %d", x, y, z);  
    }  
  end  
endmodule
```

☐ value of x is x
x is x y is y z is z

☐ value of x is 2
x is y y is z z is 10

☐ value of x is 4

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DELL

If the propagation delay of RTL-NOR gate is 500ns, then is the RTL-NOR gate can work at the speed of 4MHz?

- ☐ It cannot work because Its propagation delay is very low (in ns)
- ☐ It can work because its Fan out is 7
- ☐ It can work because its Fan in is 7
- ☐ It cannot work because Its propagation delay is very High (in ns)

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Find the impulse response $h(n)$ of a low pass FIR filter having cut-off frequency of $\frac{\pi}{2}$ rad/sample, filter should have the length of 17 with linear phase of 4.

☐ $h(n) = \frac{1}{17} \left\{ 1 + 2 \sum_{k=1}^4 \text{Re} \left[e^{-j2nk(8-n)/17} \right] \right\}$

☐ $h(n) = \frac{1}{17} \left\{ 1 + 2 \sum_{k=1}^4 \text{Re} \left[e^{j2nk(8-n)/17} \right] \right\}$

☐ $h(n) = \frac{1}{17} \left\{ 1 - 2 \sum_{k=1}^4 \text{Re} \left[e^{-j2nk(8-n)/17} \right] \right\}$

☐ $h(n) = \frac{1}{17} \left\{ 1 + 2 \sum_{k=1}^4 \text{Re} \left[e^{-j2nk(8-n)} \right] \right\}$

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DELL

A goods train traveling from station A to B meets with an accident one hour after starting. After a stoppage of 30 minutes there, due to damage it now proceeds at 80% of normal speed and arrives 2 hours late. Had the accident happened 80 km further on, it would have reached one - and - a - half hours earlier without any stoppage. What is the original speed of the train?

☐ 25 km/hr☒ 22.5 km/hr☐ 20 km/hr☐ 30 km/hr[Previous Question](#)[Next Question](#)

Mark(s): 1

SECTION: Problem Solving

A goods train traveling from station A to B meets with an accident one hour after starting. After a stoppage of 30 minutes there, due to damage it now proceeds at 80% of normal speed and arrives 2 hours late. Had the accident happened 80 km further on, it would have reached one - and - a - half hours earlier without any stoppage. What is the original speed of the train?

☐ 25 km/hr

☒ 22.5 km/hr

☐ 20 km/hr

☐ 30 km/hr

Previous Question

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DELL

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☐ 25 km/hr

☐ 22.5 km/hr

☐ 20 km/hr

☐ 30 km/hr

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DELL

Find the impulse response $h(n)$ of a low pass FIR filter having cut-off frequency of $\frac{\pi}{2}$ rad/sample, filter should have the length of 17 with linear phase of 4.

☐ $h(n) = \frac{1}{17} \left\{ 1 + 2 \sum_{k=1}^4 \text{Re} \left[e^{-j2\pi k (N-n)/17} \right] \right\}$

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☐ $h(n) = \frac{1}{17} \left\{ 1 - 2 \sum_{k=1}^4 \text{Re} \left[e^{-j2\pi k (N-n)/17} \right] \right\}$

☐ $h(n) = \frac{1}{17} \left\{ 1 + 2 \sum_{k=1}^4 \text{Re} \left[e^{-j2\pi k (N-n)} \right] \right\}$

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QUESTION: 34 GROUP: Hardware SECTION: Microprocessors and Microcontrollers
Mark(s): 1

Which of the following machine execute the instructions by microprogram?

- I) RISC
- II) CISC
- III) Microchip

☐ Only (I)

☐ Only (I) and (II)

☐ Only (II)

☐ (I), (II) and (III)

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End Test

DELL

At what temperature, a germanium semiconductor having energy gap of 0.72 eV will become conductive if electron-hole-pair generation takes place only when the operating temperature is equal or greater than the temperature corresponding to energy gap.

☐ 53000 °C

☒ 53 °C

☐ 1300 °C

☐ 5300 °C

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End Test

QUESTION 32

Mark(s): 1

GROUP: Hardware

SECTION: Microprocessors and Microcontrollers

Which one of the given option is the correct set of logic instructions?

☐ ADDC, DEC, INC, DIV☐ MOV, PUSH, POP, XCH☐ CLR, SETB, ANL, RRC☐ AJMP, ACALL, RET, JNB[Previous Question](#)[Next Question](#)

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QUESTION 41 GROUP: Hardware SECTION: Digital Electronics
Mark(s): 1

TIME LEFT 00:40:04

A 4-bit left shift register shifts by 1 bit for every clock pulse. The higher two bits are ORed and the output is fed to a T FF and the output of T FF is fed to the LSR. If the initial contents of the register are 1011, what are the contents of the register after 4 clock pulses? Initially the FF is reset.

☐ 1 0 1 1☐ 0 0 0 1☒ 1 0 1 0☐ 1 1 1 1[Previous Question](#)[Next Question](#)

DELL